

# PLATE TECTONICS

Numerous features and events at Earth's surface, from volcanic activity to earthquakes and mountain-chain formation, can be explained on the basis that our planet's outer shell is split into fragments called tectonic plates, which slowly move around on the surface.

## Earth's plates

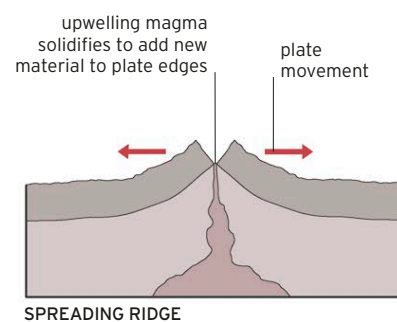
Earth's crust is fused to the top layer of the underlying mantle (see p.10), forming a shell-like structure called the lithosphere. The lithosphere is split into several chunks, called tectonic plates, which slowly move around on Earth's surface, driven by convection in the mantle. There are seven large and numerous medium-sized and smaller plates. As they move, Earth's continents are slowly shuffled around. The rate of movement is small, but over tens or hundreds of millions of years, it can produce major rearrangements.

During Earth's long history, continents carried by different plates have collided from time to time, pushing up mountains, or combined to form supercontinents.

Conversely, large landmasses have sometimes split into smaller chunks in a process called rifting.

## Plate boundaries

The science of plate tectonics largely revolves around the study of what happens near the edges of plates as they move relative to each other. These plate boundaries



## TECTONIC PLATES

Earth's plates have irregular shapes and a wide range of sizes. They cover Earth's surface like a jigsaw puzzle. Here, several plates are visible (their boundaries highlighted in orange).

## DIVERGENT BOUNDARY

At a divergent boundary, plates move apart (top). This type of boundary can be seen at the Silfra fissure in Iceland (bottom), where the Eurasian and North American Plates are separating.